

Increasing female attendance and the Geo-INQUIRE training program

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Infographic:
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What is Geo-INQUIRE?

Fabrice Cotton (GFZ, project coordinator):

“Since 2022, researchers from 51 European institutions have been collaborating on Geo-INQUIRE, a multidisciplinary Horizon Europe project. This initiative aims to enhance, provide access to, and integrate key datasets, big data streams, and High-Performance Computing (HPC) tools critical for studying temporal variations in the Solid Earth, forecasting multi-hazards and analyzing interactions between the solid Earth and its surrounding environments, including the ocean and atmosphere. Geo-INQUIRE seeks to overcome cross-domain barriers, by leveraging cutting-edge data management techniques, advanced modeling and simulation methods, developments in AI and big data, and the extension of existing data infrastructures.”

**Gender-balance is important for every project!
Set clear and ambitious goals right from the start**

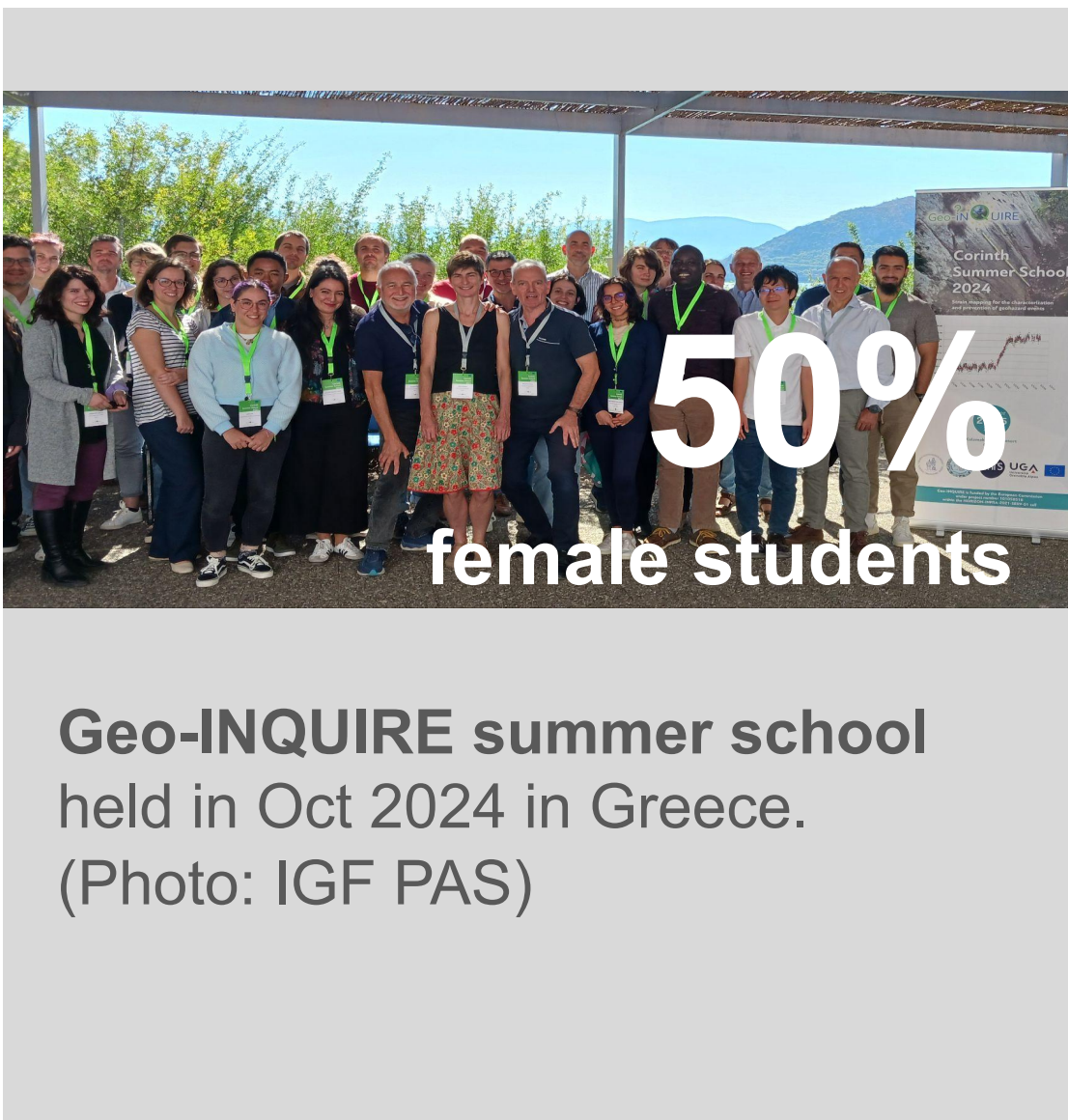
Measures taken:

- **Find good role models**
- **Increase female visibility**
- **Monitor goals**, implement countermeasures if necessary
- **Clearly state that female participation is desired**
- **Vacancies are advertised inclusively**
- **Provide asynchronous options** (recordings, slides, ...)
- **Provide daycare** if necessary
- **“Female Friendly” Environment** (open discussions, encourage questions, meet regularly)
- **“Family Friendly” Environment** (respect weekends & holidays, schedule meetings and realistic deadlines early)

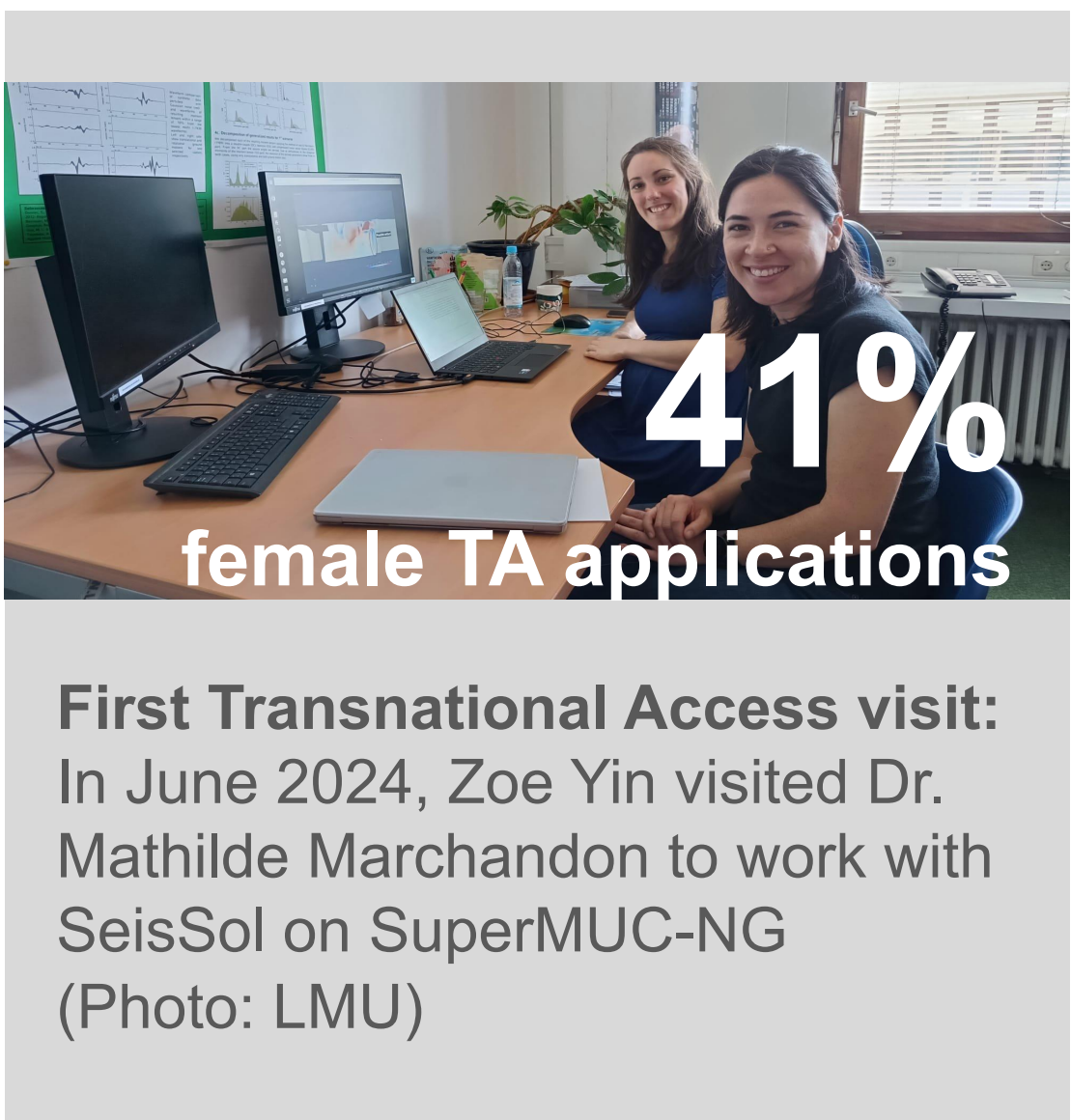
Geo-INQUIRE offers

Geo-INQUIRE strives for gender-balance in all its offers:

- **Summer Schools (50% female attendees)**, next in Catania (Oct. 2025)
- **Virtual Access (VA)** open to everyone (data products, catalogues, software, workflows, etc.)
- **Transnational Access (TA) (41%)** regular calls to visit hosting sites, receive training-through-research and HPC computing budget (e.g., working on HPC workflows from ChEESe). Targeted at advanced PhD students or Postdocs.
- **Training and Workshops (43%)** program to promote and facilitate access to many VAs and TAs. Trainings are usually online and free to attend for everybody. Workshops are on-site with limited seats & a selection process.
- **Personalized training (57%)** travel funds to work on your own project and receive training from the hosting site (targeted at early career researchers).
- **Recordings and material** from trainings and workshops asynchronously available at www.geo-inquire.eu



Geo-INQUIRE summer school
held in Oct 2024 in Greece.
(Photo: IGF PAS)



First Transnational Access visit:
In June 2024, Zoe Yin visited Dr. Mathilde Marchandon to work with SeisSol on SuperMUC-NG
(Photo: LMU)



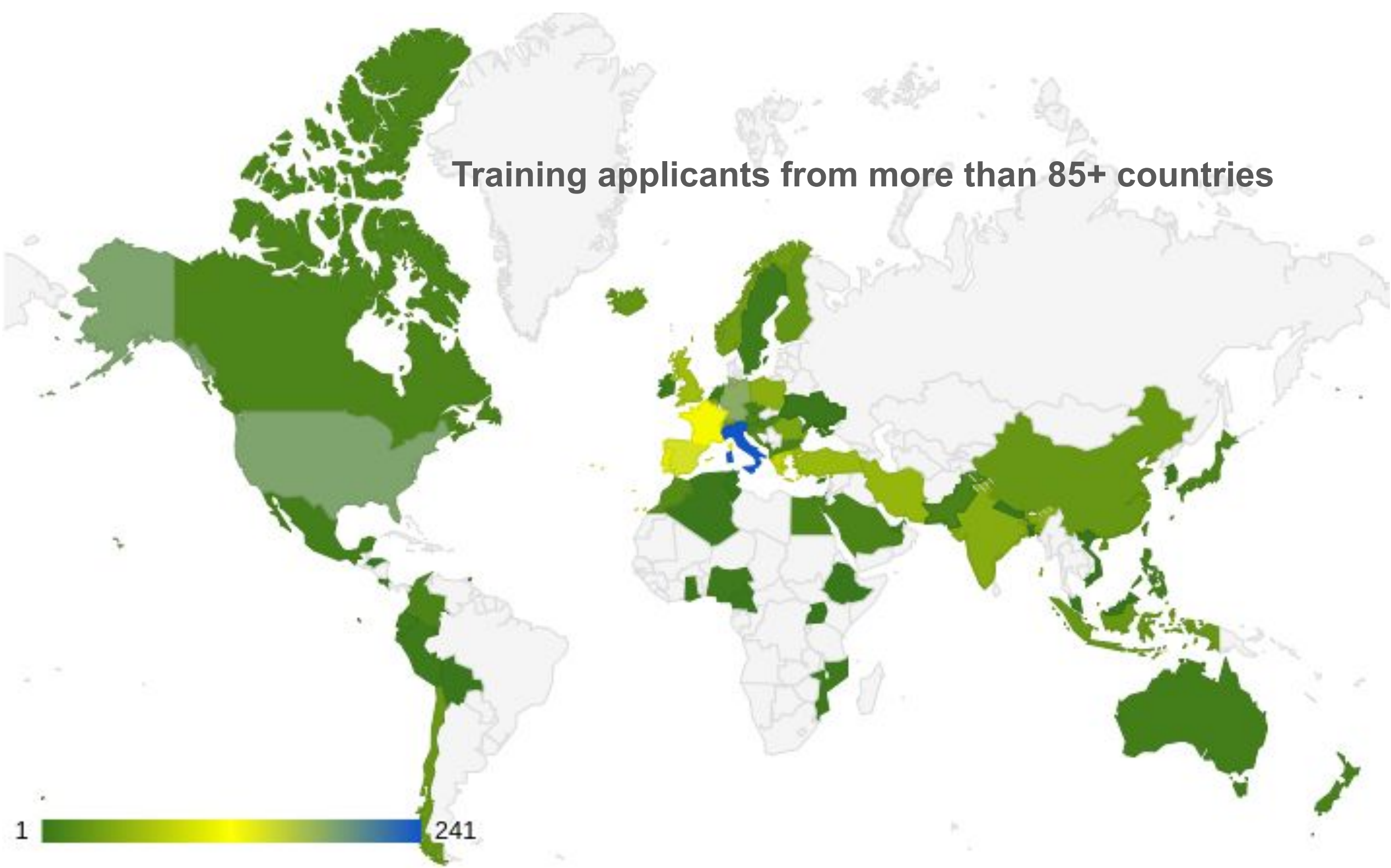
Geo-INQUIRE-Workshop on Simulation Data Lakes and Earthquake Ground Motions
Recordings and material available on the Geo-INQUIRE website
(Photo: GFZ)

The link between HPC and Geo-INQUIRE

On the **“HPC software”** side Geo-INQUIRE offers many ChEESE codes from Seismology, Volcanology and Tsunami computation as VAs and several ChEESE workflows through TAs. If a software TA is granted, applicants will also receive a computing budget on European Tier-0 or Tier-1 machines (**Leonardo@CINECA**, **SuperMUC-NG@LRZ** or **MareNostrum5@BSC**).

“Big Data”: Geo-INQUIRE offers data from mobile phones, GNSS data from satellites and distributed acoustic sensing (DAS) data from fibre optic cables.

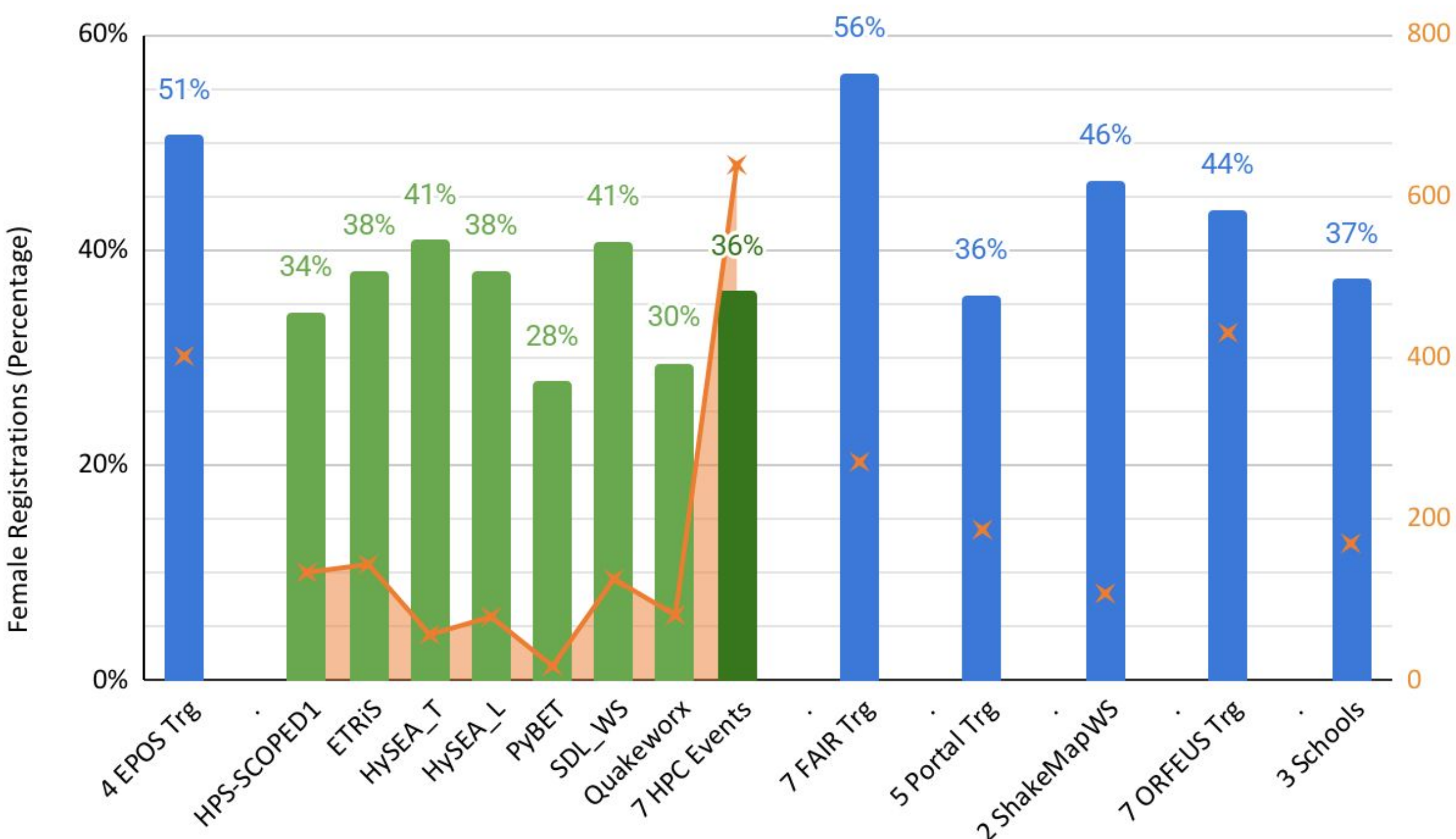
Simulation Data Lake (SDL): Stores data from HPC simulation runs but also from TAs. The SDL is **part of CINECA's infrastructure for Leonardo** and allows a tight connection between this Tier-0 machine and simulation data e.g., for analysis and visualization.



Acknowledgements

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Gender Balance in Geo-INQUIRE Training Events



Geo-INQUIRE hosted 30+ events with more than 2000 attendees. Here numbers for the 35 events hosted until May 2025 are given. We show details for the 7 HPC events and grouped results for other events to show that even for HPC (~650 participants) we could reach an average of 36% female participation.

HPC-Events:
HPS: High Performance Seismology
ETRS: European Tsunami Risk Service
HySEA_T: Tsunami-HySEA
HySEA_L: Landslide-HySEA
PyBET: Probabilistic Tsunami Hazard Ass.
SDL_WS: Simulation Data Lake Workshop
Quakework: Seismology Portal Kickoff

Legend:
bars (left axis): Percentage of Female registrations:
• green bars: HPC events
• light-green bar: Summary of HPC ev.ts
• blue bars: non-HPC events
orange stars&lines (right axis): Number of total registrations

Conclusions

Geo-INQUIRE, an interdisciplinary Geosciences project on the boundaries of Earth Science, HPC and Big Data incorporated several measures to increase female participation in all their offers. Data from their training program, which attracted in total more than **2,000 participants (from 85+ countries)** for the **30+ training events from October 2022 to May 2025** shows that, contrary to many expectations, their gender balance goal of striving for **40% female engagement** could be reached. Albeit women being underrepresented in both Earth Science and HPC, the participation rate of women per event fell usually between a third and two thirds(!). The average percentage of female participants in all Geo-INQUIRE **training events is 43%**. Two thirds of the successful applicants to **personalized training** and half of those for the first **summer school** were women. About 40% of the **TA applications** and admitted projects are from female PIs. We have shown that the measures taken have been very successful and would like to encourage other HPC projects to follow similar paths and put focus on gender balance and diversity from the start on.



see <https://www.geo-inquire.eu/> for information on Virtual and Transnation Access, Training & Workshops (incl. material and recordings), Personalized Training & Summer Schools

